# Due: August 18, 2026

1. Given $q = \frac{(v+2)^3-8}{v}$ and $(v \neq 0)$, find:

   a) $\lim_{v \to 0} q$

   b) $\lim_{v \to 2} q$

2. Compute the following limits:

   a) $\lim_{x \to \infty} \dfrac{x}{x^2+1}$

   b) $\lim_{x \to \infty} \dfrac{x^2}{x^2+1}$

   c) $\lim_{x \to \infty} \dfrac{x^4}{x^2+1}$

   d) $\lim_{x \to 8+} \dfrac{5}{x-8}$

   e) $\lim_{x \to 8-} \dfrac{-5}{x-8}$

3. Given the function $y = 3x + 5$

   a) Find the difference quotient as a function of $x + \Delta x$.

   b) Find the derivative $dy/dx$.

   c) Find $f'(1)$ and $f'(3)$.

4. Given the function $y = 5x^2 - 4x$

   a) Find the difference quotient as a function of $x + \Delta x$.

   b) Find the derivative $dy/dx$.

   c) Find $f'(2)$ and $f'(3)$.

5. Compute the derivatives for the following functions by use of the power rule:

   a) $f(x) = -6x^3 + 12x^2 - 4x + 7$

   b) $f(x) = \frac{1}{10}x^{10} + \frac{1}{9}x^9 - \frac{1}{8}x^8$

   c) $f(x) = (3x+5)^2$ (expand first)

d) $f(x) = 100x^2 + \frac{1}{100}x$

6. Compute the derivatives $y'$ two ways: product rule and expanding first:

   a) $y = (x^2+1)(x^3+1)$
   b) $y = (x-1)(x^5+x^4+x^3+x^2+x+1)$
   c) $y = (x^3+1)(x^2+1)(x+1)$

7. Compute the following derivatives by using the quotient rule:

   a) $y = \dfrac{5x+6}{x^2+1}$

   b) $y = \dfrac{x^3-x}{x^5-x^3+1}$

8. Compute the following derivatives by using the quotient rule and the product rule:

   a) $y = \dfrac{(x^3-2x^2+7x+1)(x^2-x+3)}{x^3+1}$

   b) $y = \dfrac{(x^2+1)(x^4+1)}{(x^3+1)(x^5+1)}$

9. Differentiate the following functions:

   a) $e^{5x}$
   b) $e^{\sqrt[3]{x}}$
   c) $(e^e)^x$
   d) $7^x$
   e) $2^{(x^2+6x-5)}$
   f) $\ln(x^4-x^2+1)$
   g) $\ln(e^x)$
   h) $\ln(2x)$